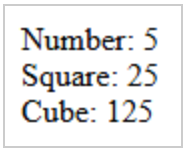
Prelab Programs: Week 5

1. Write JS code using an arrow function to find the square and cube of a given number.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Square and Cube</title>
</head>
<body>
    <script>
        let n = 5;
        const squareCube = (num) => {
            return { square: num * num, cube: num * num * num };
        };
        let result = squareCube(n);
        document.write(`Number: ${n}<br>`);
        document.write(`Square: ${result.square}<br>`);
        document.write(`Cube: ${result.cube}`);
    </script>
</body>
</html>
```

Output:



2. Write JS code to demonstrate a callback function that adds two numbers and displays the result through the callback.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Callback Function</title>
</head>
<body>
    <script>
        function add(a, b, callback) {
            let sum = a + b;
            callback(sum);
        }

        function displayResult(result) {
            document.write(`The sum is: ${result}`);
        }

        add(5, 10, displayResult);
    </script>
</body>
</html>
```

Output:

The sum is: 15

3. Write JS code using a Promise that resolves after 2 seconds with a success message and rejects if a given number is negative.

Code:

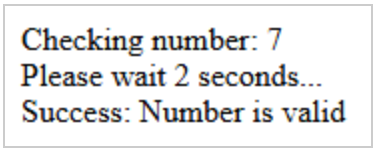
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Promise Example</title>
</head>
<body>
    <div id="output"></div>
    <script>
        let n = 7;
        let output = document.getElementById("output");

        function checkNumber(num) {
            return new Promise((resolve, reject) => {
                setTimeout(() => {
                    if (num < 0) {
                        reject("Error: Number is negative");
                    } else {
                        resolve("Success: Number is valid");
                    }
                }, 2000);
            });
        }

        output.innerHTML = `Checking number: ${n}<br>Please wait 2 seconds...`;

        checkNumber(n)
            .then((message) => {
                output.innerHTML += `<br>${message}`;
            })
            .catch((error) => {
                output.innerHTML += `<br>${error}`;
            });
    </script>
</body>
</html>
```

Output:



4. Write JS code using async/await to simulate fetching data (using setTimeout inside a Promise) and print it after a delay of 3 seconds.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Async Await Example</title>
</head>
<body>
    <div id="output"></div>
    <script>
        let output = document.getElementById("output");

        function fetchData() {
            return new Promise((resolve) => {
                setTimeout(() => {
                    resolve("Data fetched successfully from server");
                }, 3000);
            });
        }

        async function showData() {
            output.innerHTML = "Fetching data, please wait 3 seconds...<br>";
            let data = await fetchData();
            output.innerHTML += data;
        }

        showData();
    </script>
</body>
</html>
```

Output:

Fetching data, please wait 3 seconds...
Data fetched successfully from server

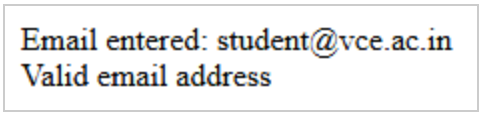
5. Write JS code using a regular expression to validate whether a given string is a valid email address.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Validate Email</title>
</head>
<body>
    <script>
        let email = "student@vce.ac.in";
        let pattern = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;

        document.write(`Email entered: ${email}<br>`);
        if (pattern.test(email)) {
            document.write("Valid email address");
        } else {
            document.write("Invalid email address");
        }
    </script>
</body>
</html>
```

Output:



6. Write JS code to change the background colour of a webpage when a button is clicked, using an event handler.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Change Background Colour</title>
</head>
<body>
    <button id="colorBtn">Change Background Colour</button>

    <script>
        let colors = ["lightpink", "lightgreen", "lightblue", "lightyellow"];
        let btn = document.getElementById("colorBtn");

        btn.addEventListener("click", function () {
            let randomColor = colors[Math.floor(Math.random() * colors.length)];
            document.body.style.backgroundColor = randomColor;
        });
    </script>
</body>
</html>
```

Output:

